INCH-POUND

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SUPERSEDING
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18 May 1982

MILITARY SPECIFICATION

LIFE PRESERVER SUPPORT PACKAGE FOR LIFE PRESERVER, MK 4

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for construction and performance for MK 4 life preserver, repair kit and spare parts and test kit for the life preserver support package (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4220

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SPECIFICATIONS

FEDERAL

V-T-285	Thread, Polyester
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stack (Container Grade), and Cut Shapes

MILITARY

MIL-L-3454	Life Preservers: Packaging of
MIL-R-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for
MIL-C-16385	Cartridge, Carbon Dioxide, Inflatable Life Preserver Use
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated
MIL-M-19595	Magnetic Effect Limits for Nonmagnetic Equipment Used in the Proximity of Magnetic Influence Ordnance
MIL-C-25369	Cylinders, Carbon Dioxide, for Inflation of Pneumatic Life Preservers
MIL-W-44049	Webbing, Textile and Tape, Textile, Polypropylene, General Purpose, Natural or in Colors

STANDARDS

FEDERAL

FED-STD-595	Colors

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

592-6383528	Life Preserver Support Package – Life Preserver, Mark 4 (Assembly and Data List)
592-6383529	Life Preserver Assembly, Yoke Type Scuba, USN Mark 4
592-6383530	Cover Assembly, Outer
592-6383532	Bladder Assembly, Inner
592-6383533	Stem Assembly, Inflator

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592-6383564	CO_2 Holder and Holder Gage Details
592-6383565	Spare Parts/Test Kit
592-6383581	Repair Kit
803-1411524	Holder CO_2 and Holder Gage Details

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

PUBLICATIONS

UNITED STATES COAST GUARD
Code of Federal Regulations, Title 46, Sub-Chapter II

160.060	Marine Buoyant Devices
160.064	Marine Buoyant Devices

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 117	Standard Method of Salt Spray (Fog) Testing; (DOD adopted)
B 194	Standard Specification for Copper-Beryllium Alloy Plate, Sheet, Strip and Rolled Bar; (DOD adopted)
B 196	Standard Specification for Copper-Beryllium Rod and Bar [Metric]; (DOD adopted)
B 197	Standard Specification for Copper-Beryllium Alloy Wire; (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this

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document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Material. The life preserver shall be constructed of materials as specified on the applicable drawings and as specified herein.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Nonmagnetic material. The life preserver shall be constructed of material which withstands the magnetic effects test of 4.9.13.

3.2.3 Mechanical inflation system.

3.2.3.1 Inflator assemblies. Each life preserver shall be provided with two assemblies consisting of two dual inflators conforming to Drawing 592-6383564. The assembly shall consist of holders for use with CO_2 cartridges conforming to type II of MIL-C-16385 or type I of MIL-C-25369. Holders shall withstand a hydrostatic pressure of 1,500 pounds per square inch (lb/in^2) gauge for 0.5 seconds.

3.2.3.2 Preserver holder assemblies. The preserver holder assemblies shall have levers and setscrews fabricated from Beryllium Copper conforming to ASTM B194. There shall be provisions for securing the actuating levers against the sides of the holders by a red retaining clip. The holder threads shall be cold formed, rolled thread, and shall be coated with a silicone grease after the holders have been assembled on the life preserver. There shall be Beryllium Copper springs made of wire conforming to ASTM B194 to provide that the piercing pins separate from the cartridges immediately after puncturing to preclude any possibility of delay in inflation by the pins remaining in and sealing the holes. The piercing pins shall be Beryllium Copper conforming to class A of ASTM B196 and B197.

3.2.3.3 Inflation lanyard. The inflation lanyard assembly shall be as detailed on Drawing 592-6383529. The lanyard assembly shall not break or fail at less than a minimum breaking strength of 50 pounds (see 4.9.15).

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3.2.3.4 Inflator stem. The holder shall be securely attached to the life preserver by means of an uncoated brass inflator stem conforming to Drawing 592-6383533. Two synthetic rubber gaskets conforming to class 2, type A, grade 80 of MIL-R-6855 shall be provided, one for each side of the holder attachment. The inflator stem shall provide a zero leakage connection between the inflation assembly and the life preserver.

3.2.4 Color. Life preserver parts and materials shall have exterior color in accordance with number 37038 of FED-STD-595.

3.2.5 Thread. Thread used for all sewing shall conform to type II, class 1, size F of V-T-285 and shall be colored to match the items sewn.

3.2.6 Bladder. The material used for the bladder shall be a polyether base, polyurethane film that can be dielectrically heat sealed to itself. The film shall be fungus resistant and contain no waxes or other ingredients which adversely affect its dielectric sealability. The film shall be of a clear color with a matte finish on both sides. The material shall have air-holding characteristics and be free of pin holes, surface imperfections, and particles of foreign matter. The film shall meet the physical requirements of table I (see 6.3).

TABLE I. *Physical requirements of polyurethane film.*

Characteristic	Requirement
Specific gravity	1.1
Hardness	80 Shore A (typical)
Brittle temperature (masland SPI cold crack)	Minus 70 degrees Fahrenheit (°F)/maximum (max.)
Tensile strength (ultimate)	3000 lb/in^2 minimum (min.)
Ultimate elongation (at break)	500 percent min.
Tear strength (graves)	400 lb/in^2 min.
Abrasion resistance: (taber) 1000 cycles, CS17 wheel; load 1000 grams Percent weight loss	0.06 percent max.

3.2.7 Outer covering. The outer covering shall conform to the requirements (see 6.3) on Drawing 592-6383530 and meet the physical requirements of table II.

TABLE II. *Physical requirements of outer covering.*

Characteristic	Requirement
Burst strength (Mullen Hydrostatic)	600 pounds min.
Tensile strength (grab method)	500 warp x 400 fill pounds min.
Elongation at break	37 x 32 percent max.
Tear (tongue method)	20 x 20 pounds min.
Puncture resistance	70 pounds min.
Tear (trapezoid method)	30 x 20 pounds min.
Blocking scale rating	No. 2

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3.2.8 **Webbing.** The webbing shall be in accordance with MIL-W-44049 and shall meet the performance requirements for webbing of Coast Guard Specifications 160.060 and 160.064 (see 6.3).

3.2.9 **Corrosion resistance.** The oral valve, relief valve, CO_2 inflation devices and all metal parts shall be made of corrosion-resistant material and shall show no evidence of corrosion after subjection to the 100-hour salt spray test of 4.9.12.

3.3 **General requirements.**

3.3.1 **Manufacture and assembly.** The manufacturing and assembly details of the life preservers shall conform to Drawing 592-6383528.

3.3.2 **Heat sealing.** A radio frequency dielectric heat sealing process shall be used in the attachment of all parts and the construction of the seams of the inner bladder. The seams of the inner bladder shall be as shown on Drawing 592-6383532. The seams shall withstand a 20 pound per inch (lb/in) peel load (see 4.9.11).

3.3.3 **Marking.** Identification marking shall be by silkscreen method using white ink. The marking shall be waterproof and cause no deterioration of fabric.

3.3.3.1 **Serial numbers.** Each life preserver shall be identified by a consecutive number (serial number) assigned by the manufacturer to cover the entire contract quantity. Serial numbers shall be located in accordance with Drawing 592-6383530.

3.3.4 **Repair kit.** Each life preserver shall be provided with a repair kit consisting of the items listed on Drawing 592-6383581.

3.3.5 **Spare parts or test kit.** Two spare parts or test kits shall be provided. Each spare parts or test kit shall consist of the items listed on Drawing 592-6383565.

3.4 **Reliability.** The life preserver shall be constructed in a manner that will ensure a 97 percent operational reliability with a 95 percent confidence level under all combinations of storage and operating environments (see 4.9.14).

3.5 **Operation.** When an inflation mechanism containing two CO_2 cartridges is released into the life preserver, the preserver shall fully inflate in not more than 5 seconds.

3.6 **Hydrostatic pressure.** The life preserver shall operate properly after being subjected to the hydrostatic pressure cycle test (see 4.9.1.1).

3.7 **Impact.** The life preserver shall withstand an impact of 18 feet per second.

3.8 **Buoyancy.** The life preserver shall support a lead weight, weighing 42 pounds, without sinking.

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3.9 Operation of oral valve.

3.9.1 During assembly. The purge ports of the oral valve shall operate properly before and after the button is depressed.

3.9.1.1 After assembly. After inflation through the oral valve, the purge ports shall prevent water from entering the inner bladder. When the preserver is inflated until the relief valve opens, the oral valve shall operate and shall not leak when the preserver is fully inflated. The oral valve shall work without difficulty and shut off securely when the preserver is deflated.

3.10 Pressure. The preserver shall withstand a pressure test of 5 lb/in^2 after the over-pressure relief valve is removed and the opening capped.

3.11 Leakage.

3.11.1 Twenty-four hours. The life preserver shall have an internal pressure, corrected for barometric pressure and temperature, of not less than 1.0 lb/in^2 after being inflated to 1.5 lb/in^2 and allowed to stand for 24 hours.

3.11.2 Six hours. The life preserver shall remain firm after being inflated until the relief valve opens and allowed to stand for 6 hours.

3.12 Operation of relief valve. The relief valve shall relieve at a pressure of 2.5 $\pm$ 0.5 lb/in^2 (see 4.9.6).

3.13 Oral inflation hose.

3.13.1 Porosity. The oral inflation hose shall show no evidence of leakage due to porosity nor weakening of the hose as revealed by excessive deformation in any portion of the hose when subjected to an internal air pressure of 15 lb/in^2 for not less than 10 seconds while submerged in water (see 4.9.9.1).

3.13.2 Tensile strength. The tensile strength of the oral inflation hose shall be not less than 50 pounds (see 4.9.9.2).

3.14 Holding strength (oral inflation assembly). The load required to separate the hose from the oral inflation valve or oral inflation elbow shall be not less than 50 pounds (see 4.9.10).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on the sample units as specified in 4.3.1. First article inspection shall consist of the examination and tests of table III. Failure of any examination or test specified shall be cause for refusal to grant first article approval.

4.3.1 Samples for first article inspection. Prior to starting regular production, the following samples shall be submitted for first article inspection:

a. One complete life preserver support package, including properly charged CO_2 cartridges (see Drawing 592-6383528).
b. Four pieces mesh material 2-3/4 inches wide by 8 inches long.
c. Two yards each of all outer covering and urethane bladder material used in life preserver and preserver cover. The name of the supplier (if different from the manufacturer) and part numbers shall be furnished with the samples.

TABLE III. *First article inspection.*[1]

Inspections	Inspection	Number of sample units inspected
Visual examination	4.6	8
Operation	4.9.1	8
Buoyancy	4.9.2	8
Operation of oral valve	4.9.3.2	8
Pressure	4.9.4	8
Leakage	4.9.5	8
Operation of relief valve	4.9.6	8

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TABLE III. *First article inspection.*[1]

Inspections	Inspection	Number of sample units inspected
Dimensional inspection (gauge) of CO_2 inflation assembly	4.9.7	22
Leakage and operation (CO_2 inflation assembly)	4.9.8	22
Oral inflation assembly holding strength	4.9.10	4
Radio Frequency (RF) heat sealing	4.9.11	1
Salt spray	4.9.12	1
Magnetic effect (tests are in addition to those required by note 1)	4.9.13	3
Reliability	4.9.14	2
Breaking strength (lanyard assembly)	4.9.15	4
Inspection of packaging	4.10	8

[1]Life preservers shall have been subjected to and passed the magnetic effect inspection of 4.9.13 before submission for first article inspection.

d. One complete inner bladder assembly.
e. Two yards each of all fastening tape used on preserver.
f. One cone (500 yards) of polyester sewing thread of the color used in the preserver.
g. Four sets of all hardware, including CO_2 inflator system, and zippers used in life preserver. The name of the supplier (if different from the manufacturer), and part number shall be furnished with the samples.
h. Four pressure relief valve assemblies and oral inflation assemblies including internal and external threaded access fittings and seal rings. The name of the supplier (if different from the manufacturer), and part number shall be furnished with the samples.

4.3.2 First article sample identification. Samples for first article inspection shall be forwarded to the contracting activity plainly identified by securely attached durable tags marked with the following information:

a. Samples for first article inspection.
b. Name of product, together with complete identifying data such as manufacturer's designation number.
c. Name of manufacturer.
d. Contract or order number.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the visual examination of 4.6, and the tests specified for groups A and B (see 4.4.1 and 4.4.2) (see 6.3).

4.4.1 Group A tests. Group A tests shall be conducted in accordance with table IV as a part of the inspection system (see 4.1.1) during the assembly of the life preserver. When specified

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(see 6.2), tests may be witnessed by the Government inspector. In addition, the contractor shall prepare RF heat sealed test samples from a test bladder and shall subject the samples to the test of 4.9.11. One bladder shall be so tested at the start of each day's production and every fiftieth bladder thereafter.

4.4.2 Group B tests. Group B tests shall be conducted as specified in table V on life preserver samples selected as specified in 4.5.3 which have passed the visual examination tests (see 4.6) and the component tests of 4.4.1. When specified (see 6.2), tests shall be witnessed by the Government inspector. Any item in the sample containing one or more defects shall not be offered for delivery.

TABLE IV. *Group A tests.*

Test	Quantity	Paragraph
Operation of oral valves	Each	4.9.3.1
Leakage – six-hour test	Each	4.9.5.2
Dimensional examination of inflation assembly	Each	4.9.7
Leakage and operation (inflation assembly)	Each	4.9.8
Oral inflation hose (porosity)	Six samples per 1000	4.9.9.1
Oral inflation hose (tensile)	Six samples per 1000	4.9.9.2
Oral inflation assembly (holding strength)	Six samples per 1000	4.9.10
RF heat sealing	1 every 50	4.9.11

TABLE V. *Group B tests.*

Test	Paragraph
Operation (CO_2 inflation)[1]	4.9.1
Buoyancy[1]	4.9.2
Operation of oral valve	4.9.3.2
Pressure	4.9.4
Leakage - 24-hour test	4.9.5.1
Operation of relief valve	4.9.6

[1]The buoyancy test specified in 4.9.2 shall be conducted when the CO_2 cartridges are expended during the operation test of 4.9.1.

4.5 Sampling procedures.

4.5.1 Lot. A lot shall consist of all preservers submitted for inspection at one time. Maximum lot size shall not exceed 250 life preservers.

4.5.2 Sampling for visual examination (major and minor defects). A random sample of life preservers shall be selected from each lot as specified in the sampling plan (table VI) for the visual examination for major and minor defects in 4.6.2 (see 6.8).

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TABLE VI. *Sampling for visual examination.*

Lot size	Sample size
Major defects	
2 – 25	3
26 – 150	8
151 – 280	13
281 – 500	20
501 – 1200	32
1201 – 3200	50
Minor defects	
2 – 15	2
16 – 90	5
91 – 150	8
151 – 280	13
281 – 500	20
501 – 1200	32
1201 – 3200	50

4.5.3 Sampling for group B tests. A random sample of life preservers shall be selected from each lot as specified in the sampling plan (table VII) for tests specified in 4.6.2. The acceptance number shall be zero for all tests (see 6.7).

TABLE VII. *Sampling for group B tests except buoyancy.*

Lot size	Sample size
2 – 15	2
16 – 25	3
26 – 90	5
91 – 150	8
151 – 280	13
281 – 500	20
501 – 1200	32
1201 – 3200	50
Sampling for group B buoyancy test	
2 – 50	2
51 – 500	3
501 – 3200	5

4.6 Visual examination. Life preservers shall be thoroughly visually examined to determine conformance to this specification and the applicable drawings.

4.6.1 For critical defects. The life preservers shall be examined 100 percent for critical defects as defined in table VIII. Life preservers containing one or more critical defects shall not be offered for delivery.

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TABLE VIII. *Classification of defects.*

Categories	Defects
Critical:	
1	Buoyancy chamber material defective; evidence of splits, tears, weak spots, patched holes, or fraying
2	Buoyancy chamber seams defective, incomplete, or not as specified; evidence of bubbles, incomplete sealing, or seams in buoyancy chamber not the specified type; width of seams less than the required minimum
3	Oral inflation valve defective or not secure in hose; hose evidences cracks, splits, bubbles, bulges, pin holes, or other deleterious effects
4	Oral inflation elbow or oral inflation hose not secure
5	Pressure relief valves missing, defective, not as required, not properly installed and located
6	Internal and external threaded access fittings defective, not properly installed, or improperly sealed
7	Holder and inflation mechanism assembly defective, not securely attached, or not as specified, lanyard not securely attached to actuating lever, lever retaining clip not staked in place
8	Failure of specified gauge to seat firmly against seating surface of holder, O-rings (for holder and piercing pin) missing, defective, or not as required (examine prior to assembly)
9	Gaskets (each side of holder attachment) missing, defective, or not as required
10	Zipper missing, defective, not as required, not properly installed and located
11	Evidence of reworking air retaining seam, or attachments of manifold, or seal ring
Major:	
101	Velcro for securing inflation hose defective or not securely attached as specified
102	Velcro for covering zipper tab defective or not securely attached as specified
103	Holder not secure; inflator cover not secure
104	Harness webbing evidences tears, splits, fraying, or weak spots
105	Harness stitching defective, incomplete, or not as specified
106	Harness hardware missing, defective, or not secure; evidence of cracks, deep pits, thin spots, or sharp surfaces
107	Holder actuating lever not safety wired as specified
108	Metal parts (holder, lever or manifold) not finished as specified; finish defective or incomplete
109	Date of manufacture marking missing, not permanent, not located as specified, illegible, or incorrect
111	Holder threads not greased as required after assembly on life preserver
Minor:	
201	Length of lanyard not as required
202	Color of life preserver parts and materials not as specified
203	Marking, manufacturer's name or trademark and identification missing, incorrect, illegible, not permanent, or not located as specified
204	Packaging not as specified
205	Stitching on outer cover not as specified

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4.6.2 For major and minor defects. Sample preservers selected in accordance with 4.5.2 shall be examined for major and minor defects (as defined in table VIII). Any sample containing one or more defects shall not be offered for delivery.

4.7 Magnetic effect levels. Assembled life preservers shall be submitted to the Navy Explosive Ordnance Disposal Technology Center (NEODTC), Attn: Code 45, Indian Head, MD 20640 for testing in accordance with MIL-M-19595 prior to acceptance.

4.8 Test conditions.

4.8.1 Atmospheric conditions. Unless otherwise specified herein all tests required by this specification shall be conducted at prevailing atmospheric pressure, a temperature of 77 ± 18 degrees Fahrenheit (°F), and a relative humidity no greater than 95 percent. When the inspections are conducted at an atmospheric pressure or temperature different from the values specified herein (see 6.2), proper corrections shall be made for the change in instrument readings. The instrument used shall contain no mercury.

4.8.2 CO_2 cartridge for test purposes. CO_2 cartridges used in conducting the tests specified in 4.9.1 shall be in accordance with type I of MIL-C-25369.

4.9 Test methods. Unless otherwise specified herein, the contracting activity shall have approval authority for all test equipment and methods used in performance of tests and inspections required herein, in order to ensure adequacy of product testing.

4.9.1 Operation (CO_2). The operation of the preserver shall be tested by releasing the CO_2 cartridges into the life preserver in the normal manner. The life preserver shall fully inflate in not more than 5 seconds.

4.9.1.1 Hydrostatic pressure. The uninflated life preserver shall be tested for normal operation immediately after passing through the following hydrostatic pressure cycle:

a. With the life preserver immersed in water, increase hydrostatic pressure at a rate equivalent to a descent of 75 feet per minute until a seawater depth of 200 ± 10 feet is reached.
b. Hold for 1 hour, then decrease pressure at a rate equivalent to an ascent of 60 feet per minute to the surface.

4.9.1.2 Impact. The uninflated life preserver shall be tested for normal operation immediately after being dropped into seawater at a velocity of at least 18 feet per second at impact.

4.9.1.3 Failure. The preserver shall be considered to have failed the operational tests if any of the following occur:

a. Failure of life preserver to inflate when lanyard is pulled; evidence of leaks through bladder, seams, inflation assembly or piercing pin, O-rings, access fittings, over-pressure

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valve, or oral inflation hose and assembly.

b. Life preserver cannot be readily inflated through the oral inflation valve, or there is evidence of leakage back through valve after inflation.

c. Relief valves fail to relieve within the specified limits when the preserver is inflated as required.

4.9.2 Buoyancy. Immediately upon completion of the operation CO_2 test (see 4.9.1), the preserver shall be attached to a lead weight, the latter weighing 42 pounds in air. The preserver and weight shall then be placed in clear fresh water. The life preserver shall support this weight without sinking.

4.9.3 Operation of oral valve. The tests specified in 4.9.3.1 and 4.9.3.2 shall be conducted as specified in 4.3 and 4.4.

4.9.3.1 During assembly. Pass air through the oral valve from the mouthpiece. The purge ports shall operate properly without the button depressed and the air shall pass through the oral valve properly after the button is depressed.

4.9.3.2 After assembly. The life preserver shall be inflated through the oral valve. The purge ports shall operate properly to prevent water from entering the inner bladder. The preserver shall be inflated until the relief valve opens. The oral valve shall operate without difficulty and after the vest is fully inflated shall not leak. The life preserver shall then be deflated through the oral valve and it shall be ascertained that the valve works without difficulty and shuts off securely.

4.9.4 Pressure. With the over-pressure valve removed and the test gauge installed on the over-pressure valve external threaded access fitting, the life preserver shall be inflated with air through the fitting on the test gauge to a pressure of 5 lb/in^2 and shall be allowed to stand for not less than 10 minutes. The preserver shall then be charged and readjusted to 5 lb/in^2 and if the pressure at the end of a further 30 minute period, minimum is less than 4.5 lb/in^2, the preserver shall be rejected.

4.9.5 Leakage. The following leakage tests shall be performed on the completed life preserver as specified.

4.9.5.1 Twenty-four hour test. With the over-pressure valve removed and the test gauge installed on the over-pressure valve external threaded access fitting, the life preserver shall be inflated with air through the oral inflation valve to a pressure of 1.5 $\pm$ 0.2 lb/in^2 all pressure is gauge unless otherwise specified and allowed to stand for not less than 15 minutes. The pressure shall then be checked and readjusted if necessary to the original pressure of 1.5 $\pm$ 0.2 lb/in^2. The temperature and barometric pressure shall be recorded at this time. At the end of 24 hours after the readjustment period, the pressure shall be measured and corrected for any changes in temperature or barometric pressure (see 4.8.1). The corrected pressure shall be not less than 1.0 lb/in^2.

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4.9.5.2 Six-hour test. The completed life preserver shall be inflated until the relief valve opens and allowed to stand for not less than 6 hours. At the end of this required period the life preserver shall be firm.

4.9.6 Operation of relief valve. The oral inflation elbow shall be removed and the test gauge installed on the oral inflation assembly threaded access fitting. The preserver shall then be inflated with air through the fitting on the test gauge to test the operation of the relief valve. Each valve shall relieve at a pressure of 2.5 ± 0.5 lb/in^2.

4.9.7 Dimensional inspection (gauge) of inflation assembly. Holder assembly (complete with O-ring) shall be dimensionally checked with a plug gauge conforming to the dimensions shown on Drawing 803-1411524. Failure of the plug gauge to seat firmly against the seating surface (where the puncturable end of the cartridge seats within the holder) shall be cause for rejection.

4.9.8 Leakage and operation (inflation assembly). Holders shall be tested for leakage under water with air at a pressure of 4.0 lb/in^2, with the actuating lever in the actuated position. (The actuated position is considered to be the position of the lever after the piercing pin has punctured the cartridge and has completely retracted. Cartridges are not required to determine the actuated position.) Holders shall be subjected to a hydrostatic pressure of 3,500 lb/in^2. Holders shall also be tested for leakage under water with the actuating lever slowly carried through one complete cycle.

4.9.9 Oral inflation hose. Before the oral inflation valve is installed in the oral inflation hose, the tests specified in 4.9.9.1 and 4.9.9.2 shall be conducted.

4.9.9.1 Porosity. The oral inflation hose shall be subjected to an internal air pressure of 15 lb/in^2 for not less than 10 seconds while submerged in water. There shall be no evidence of air leakage due to porosity. The shall be no evidence of weakening of the hose as revealed by excessive deformation in any portion of the hose.

4.9.9.2 Tensile. The ends of the oral inflation hose shall be placed in the jaws of a test machine. The rate of load application shall be 10 ± 2.5 inches per minute. The tensile strength shall be not less than 50 pounds.

4.9.10 Oral inflation assembly holding strength. This test shall be conducted after the oral inflation valve and oral inflation elbow have been attached and secured to the oral inflation hose. The elbow and valve shall be placed in the jaws of a test machine. The rate of load application shall be 10 ± 2.5 inches per minute. The load to separate the hose from the oral inflation valve or oral inflation elbow shall be not less than 50 pounds.

4.9.11 RF dielectric heat sealing. Fifteen inspection samples 1 inch wide by 2 inches long shall be prepared from the test bladder as shown on Drawing 592-6383532. Each inspection sample shall be placed in the jaws of a test machine so that the applied load will peel the seam. The rate of load application shall be 7.5 ± 2.5 inches per minute. The peel strength of each inspection sample shall be not less than 20 lb/in.

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4.9.12 Salt spray. A 100-hour salt spray test shall be conducted on a completed life preserver in accordance with ASTM B 117. The life preserver shall be tested for normal operation at the conclusion of the test.

4.9.13 Magnetic effect test. Prior to first article inspection, the following magnetic effect tests shall be conducted on hardware and complete life preservers:

a. Hardware to be used in the fabrication of completed life preservers shall be submitted to NEODTC, for testing in accordance with MIL-M-19595, prior to assembly into the preserver.
b. When only replacement hardware is acquired for life preservers, the hardware shall be submitted for testing as specified in 4.9.13(a).

4.9.14 Reliability. The life preserver shall be inflated until the relief valve opens for 100 cycles at the surface (see 3.4).

4.9.15 Breaking strength (lanyard assembly). The breaking strength of the lanyard assembly shall be determined by securing the detonator body to the upper jaw of a test machine and the pull knob in the bottom jaw. The portion of the detonator body containing the cord shall be between the upper and lower jaws. The jaws shall be separated until failure occurs. The rate of load application shall be 10 ± 2.5 inches per minute.

4.10 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Packaging requirements. Life preservers shall be preserved level A or C, packed level A, B, or C as specified (see 6.2) and marked in accordance with MIL-L-3454 and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2). In addition, for Navy acquisitions, the following applies:

a. Navy shipboard stowage fire-retardant requirements.

(1) *Lumber and plywood.* When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container construction members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

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Levels A and B – Type II – weather resistant
Category 1 – general use

Level C – Type I – non-weather resistant
Category 1 – general use

(2) *Fiberboard.* When specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The life preserver covered by this specification is intended for use as an emergency equipment by SCUBA personnel while operating under water at depths to 200 feet.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Whether acquisition is for life preservers support packages or life preservers only (see 1.1).
d. When first article inspection is required (see 3.1).
e. When group A and B testing is to be witnessed by a Government inspector (see 4.4.1 and 4.4.2).
f. When inspections may be conducted at an atmospheric pressure or temperature different from the values specified herein (see 4.8.1).
g. Level of preservation and packing required (see 5.1).
h. Bar codes and acquisition options (see 5.1).
i. When fire-retardant materials are not required (see 5.1(a)).
j. When fiberboard shall meet the flamespread and specific optic density requirements of PPP-F-320 (see 5.1(a)).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference paragraph	DID number	DID title	Suggested tailoring
3.2.6, 3.2.7, 3.2.8	DI-E-2121	Certificate of Compliance	–
4.4	DI-T-2072	Reports, Test	–

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 Quality. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Suggested accept/reject criteria for group B tests. The accept/reject criteria listed in table IX is suggested for use after the life preservers have undergone the group B tests of 4.4.2.

6.8 Suggested accept/reject criteria for visual examination. The accept/reject criteria listed in table X is suggested for use after the life preservers have undergone the visual examination of 4.6 and any defects that may have been revealed have been classified as specified in 4.6.1 and 4.6.2.

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TABLE IX. *Suggested accept/reject criteria for group B tests except buoyancy.*

Lot size	Sample size	Accept	Reject[1, 2]
2 – 15	2	0	1
16 – 25	3	0	1
26 – 90	5	0	1
91 – 150	8	0	1
151 – 280	13	0	1
281 – 500	20	0	1
501 – 1200	32	0	1
1201 – 3200	50	0	1
Suggested accept/reject criteria for the group B buoyancy test.			
20 – 50	2	0	1
51 – 500	3	0	1
501 – 3200	5	0	1

[1]All defective items must be replaced with acceptable items prior to lot acceptance.
[2]Inspect sample size until reject criteria is reached.

TABLE X. *Suggested accept/reject criteria for visual examination.*

Lot size	Sample size	Accept	Reject[1, 2]
Major defects			
2 – 25	3	0	1
26 – 150	8	1	2
151 – 280	13	1	2
281 – 500	20	2	3
501 – 1200	32	3	4
1201 – 3200	50	5	6
Minor defects			
2 – 15	2	0	1
16 – 90	5	1	2
91 – 150	8	2	3
151 – 280	13	3	4
281 – 500	20	5	6
501 – 1200	32	7	8
1201 – 3200	50	10	11

[1]All defective items must be replaced with acceptable items prior to lot acceptance.
[2]Inspect sample size until reject criteria is reached.

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6.9 Subject term (key word) listing.

Buoyancy
Heat sealing
Inflator assemblies
Inflation lanyard
Inflator stem
Mechanical inflation system

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4220-N348)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-24611A(SH)	30 November 1990

3. DOCUMENT TITLE
LIFE PRESERVER SUPPORT PACKAGE FOR LIFE PRESERVER, MK 4

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact(TPOC): R. Kumar PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: (703) 602-3474	(2) AUTOVON 332-3474
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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